

Supplemental Online Content

SUNRRISE Trial Study Group. Negative pressure dressings to prevent surgical site infection after emergency laparotomy. *JAMA*. Published online January 27, 2025. doi:10.1001/jama.2024.24764

eTable 1. Adherence to allocation

eTable 2. SF-12

eTable 3. EQ5D-5L EuroQol score

eTable 4. EQ5D-5L Health score

eTable 5a. Wound complication type as graded by the Clavien-Dindo scale in the iNPWT arm

eTable 5b. Wound complication type as graded by the Clavien-Dindo scale in the control arm

eTable 6. SSI by timepoint

eFigure 1. PRECIS-2 assessment of SUNRRISE protocol

eFigure 2. Definition of CDC criteria

eFigure 3. Sensitivity analysis forest plot

eFigure 4. Tipping point analysis

This supplemental material has been provided by the authors to give readers additional information about their work.

Table of Contents

Supplementary eTable 1: Adherence to allocation	3
Supplementary eTable 2: SF-12	3
Supplementary eTable 3: EQ5D-5L EuroQol score	4
Supplementary eTable 4: EQ5D-5L Health score	4
Supplementary eTable 5a: Wound complication type as graded by the Clavien-Dindo scale in the iNPWT arm	5
Supplementary eTable 5b: Wound complication type as graded by the Clavien-Dindo scale in the control arm	5
Supplementary eTable 6: SSI by timepoint	6
Supplementary eFigure 1. PRECIS-2 assessment of SUNRRISE protocol	7
Supplementary eFigure 2. Definition of CDC criteria	7
Supplementary eFigure 3. Sensitivity analysis forest plot	8
Supplementary eFigure 4. Tipping point analysis	9

Supplementary eTable 1: Adherence to allocation

Adherence to allocation intervention	iNPWT (N=411)	Surgeon's preference (N=410)	Total (N=821)
Received the allocated randomised intervention?			
No	6 (2%)	8 (2%)	14 (2%)
Yes	404 (98%)	402 (98%)	806 (98%)
Unable ^a	1 (<1%)	0 (0%)	1 (<1%)
What was the dressing applied to the laparotomy wound?^b			
SUNPD	404	8	412
Conventional occlusive dressing	4	353	357
Skin glue	3	50	53
No dressing	0	1	1
Other	0	9	9

a-One participant had no intervention information ticked on the CRF, so unable to compute adherence for this participant

b-Not mutually exclusive; participants may have had more than one intervention

Supplementary eTable 2: SF-12

SF12	Statistic	Time point	iNPWT	Surgeon's preference	Total	Mean diff (95% CI) P-value	Treatment x time interaction P-value
Physical component score (PCS)	N	Baseline	306	297	603		P=0.44
	Mean [SD]		39.1 [11.7]	39.5 [11.3]	39.3 [11.5]		
	Min - Max		12.7 - 66.9	13.4 - 62.1	12.7 - 66.9		
	N	Day 7	248	232	480	0.130	
	Mean [SD]		30.4 [9.8]	30.5 [9.5]	30.5 [9.6]	(-1.755 - 2.014)	
	Min - Max		8.7 - 66.8	7.9 - 62.1	7.9 - 66.8	P=0.89	
Mental component score (MCS)	N	Day 30	235	211	446	-0.861	P=0.84
	Mean [SD]		36.1 [9.8]	37.2 [10.2]	36.6 [10]	(-2.829 - 1.106)	
	Min - Max		9.7 - 60.7	15.1 - 57.8	9.7 - 60.7	P=0.39	
	N	Baseline	306	297	603		
	Mean [SD]		44.8 [13]	45.5 [13.1]	45.1 [13]		
	Min - Max		7.7 - 70.6	10.6 - 70.6	7.7 - 70.6		
Mental component score (MCS)	N	Day 7	248	232	480	-2.190	P=0.84
	Mean [SD]		40.6 [13.2]	42.8 [12.4]	41.7 [12.9]	(-4.470 - 0.088)	
	Min - Max		8.1 - 71.2	8.5 - 70.6	8.1 - 71.2	P=0.06	
	N	Day 30	235	211	446	-1.904	
	Mean [SD]		46.9 [11.8]	47.7 [12]	47.3 [11.9]	(-4.276 - 0.469)	
	Min - Max		11.5 - 69.7	12.4 - 73	11.5 - 73	P=0.12	

Analysis adjusted for all minimisation variables with centre included as a random effect and also a treatment by time interaction term included in the model

Scores range from 0-100 with higher scores indicating better outcomes

Surgeon's preference arm is the reference category and so values greater than 0 for the mean difference indicates better outcomes for iNPWT arm

Supplementary eTable 3: EQ5D-5L EuroQol score

EQ5D-5L	Statistic	Time point	iNPWT	Surgeon's preference	Total	Mean diff (95% CI) P-value	Treatment x time interaction P-value
EuroQol score	N	Baseline	345	339	684		P=0.10
	Mean [SD]		0.57 [0.35]	0.55 [0.35]	0.56 [0.35]		
	Min - Max		-0.59 - 1	-0.59 - 1	-0.59 - 1		
	N	Day 7	292	283	575	-0.057 (-0.104 - 0.010)	
	Mean [SD]		0.44 [0.32]	0.49 [0.30]	0.46 [0.31]	P=0.02	
	Min - Max		-0.41 - 1	-0.44 - 1	-0.44 - 1		
	N	Day 14	233	238	471	-0.023 (-0.072 - 0.027)	
	Mean [SD]		0.56 [0.26]	0.56 [0.29]	0.56 [0.28]	P=0.37	
	Min - Max		-0.15 - 1	-0.24 - 1	-0.24 - 1		
	N	Day 21	229	231	460	-0.017 (-0.067 - 0.032)	
	Mean [SD]		0.60 [0.28]	0.61 [0.28]	0.60 [0.28]	P=0.49	
	Min - Max		-0.41 - 1	-0.22 - 1	-0.41 - 1		
	N	Day 30	289	264	553	-0.007 (-0.054 - 0.040)	
	Mean [SD]		0.66 [0.27]	0.65 [0.29]	0.65 [0.28]	P=0.78	
	Min - Max		-0.33 - 1	-0.17 - 1	-0.33 - 1		

Analysis adjusted for all minimisation variables with centre included as a random effect and also a treatment by time interaction term included in the model

Scores range from -0.594-1 with higher scores indicating better outcomes

Surgeon's preference arm is the reference category and so values greater than 0 for the mean difference indicates better outcomes for INPWT arm

Supplementary eTable 4: EQ5D-5L Health score

EQ5D-5L	Statistic	Time point	iNPWT	Surgeon's preference	Total	Mean diff (95% CI) P-value	Treatment x time interaction P-value
Health score	N	Baseline	340	336	676		P=0.22
	Mean [SD]		46.9 [26.6]	47.9 [26.8]	47.4 [26.7]		
	Min - Max		0 - 100	0 - 100	0 - 100		
	N	Day 7	289	278	567	1.084 (-2.389 - 4.557)	
	Mean [SD]		54.9 [23]	54.3 [21.6]	54.6 [22.3]	P=0.54	
	Min - Max		0 - 100	0 - 100	0 - 100		
	N	Day 14	231	226	457	-0.668 (-4.379 - 3.042)	
	Mean [SD]		60.8 [20.1]	60.8 [19.5]	60.8 [19.8]	P=0.72	
	Min - Max		0 - 100	0 - 95	0 - 100		
	N	Day 21	222	215	437	-2.164 (-5.926 - 1.597)	
	Mean [SD]		65.5 [20.1]	66.6 [19.1]	66 [19.6]	P=0.26	
	Min - Max		5 - 100	0 - 100	0 - 100		
	N	Day 30	281	250	531	-2.010 (-5.559 - 1.540)	
	Mean [SD]		71.4 [19.1]	71.7 [18.7]	71.5 [18.9]	P=0.27	
	Min - Max		10 - 100	0 - 100	0 - 100		

Analysis adjusted for all minimisation variables with centre included as a random effect and also a treatment by time interaction term included in the model

Scores range from 0-100 with higher scores indicating better outcomes

Surgeon's preference arm is the reference category and so values greater than 0 for the mean difference indicates better outcomes for INPWT arm

Supplementary eTable 5a: Wound complication type as graded by the Clavien-Dindo scale in the iNPWT arm

Wound complication type^a	Grade I (None/conservative management or on ward intervention)	Grade II (Antibiotic drug treatment)	Grade III (Radiological or Surgical intervention)	Grade IV (ITU admission)	Grade V (Death)	Total
Granuloma	7	1	0	0	0	8
Seroma	9	0	1	0	0	10
Haematoma	7	0	0	0	0	7
Dehiscence	34	14	3	0	0	51
Skin reaction	5	0	0	0	0	5
Other ^b	2	1	0	0	0	3
Total	64	16	4	0	0	84

a-Some participants had more than one wound complication type and so these are counted as separately

b-Other complications: 1. Retained staple, 2. Sinus, 3. Missing

Supplementary eTable 5b: Wound complication type as graded by the Clavien-Dindo scale in the control arm

Wound complication type^a	Grade I (None/conservative management or on ward intervention)	Grade II (Antibiotic drug treatment)	Grade III (Radiological or Surgical intervention)	Grade IV (ITU admission)	Grade V (Death)	Total
Granuloma	7	0	0	0	0	7
Seroma	12	3	1	0	0	16
Haematoma	2	0	0	0	0	2
Dehiscence	32	13	5	0	0	50
Skin reaction	2	0	0	0	0	2
Other ^b	4	1	1	0	0	6
Total	59	17	7	0	0	83

a-Some participants had more than one wound complication type and so these are counted as separately

b-Other complications: 1. Intramuscular left iliacus muscle haematoma, 2. Prominent stitch below wound, 3. Sore on one stitch, 4. Small fat collection, 5. Abdominal compartment syndrome, 6. Entero-atmospheric fistula

Supplementary eTable 6: SSI by timepoint

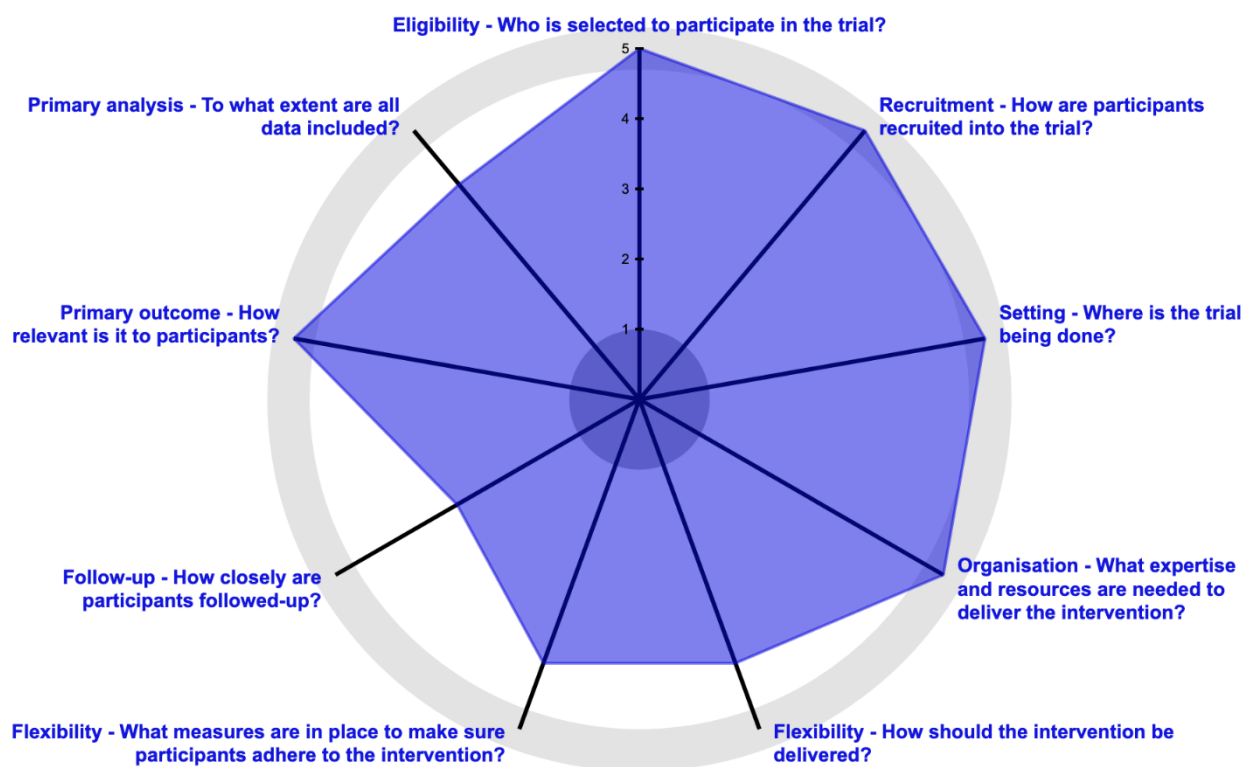
SSI by timepoint	iNPWT	Surgeon's preference
N of participants with SSI reported at Day 7 (or discharge)	37	43
N of participants with SSI reported at Return to Theatre	4	7
N of participants with SSI reported from Patient Diary	38	25
N of participants with SSI reported at Day 30	109	107
<i>N of participants with SSI reported from Day 7 or Return to theatre only</i>	<i>0</i>	<i>1</i>
<i>N of participants with SSI reported from Patient Diary only</i>	<i>3</i>	<i>0</i>

If SSI was reported at Day 7/discharge or when patient returned to theatre, if they later completed Day 30 assessment then the SSI was reported at this timepoint also.

If patient did not complete Day 30 assessment but reported SSI at earlier time-points (i.e. at Day 7/discharge or when patient returned to theatre or from patient diary), in these instances the SSI was counted towards primary analysis.

These such instances are reported in *italic text* in table above.

Supplementary eFigure 1. PRECIS-2 assessment of SUNRRISE protocol



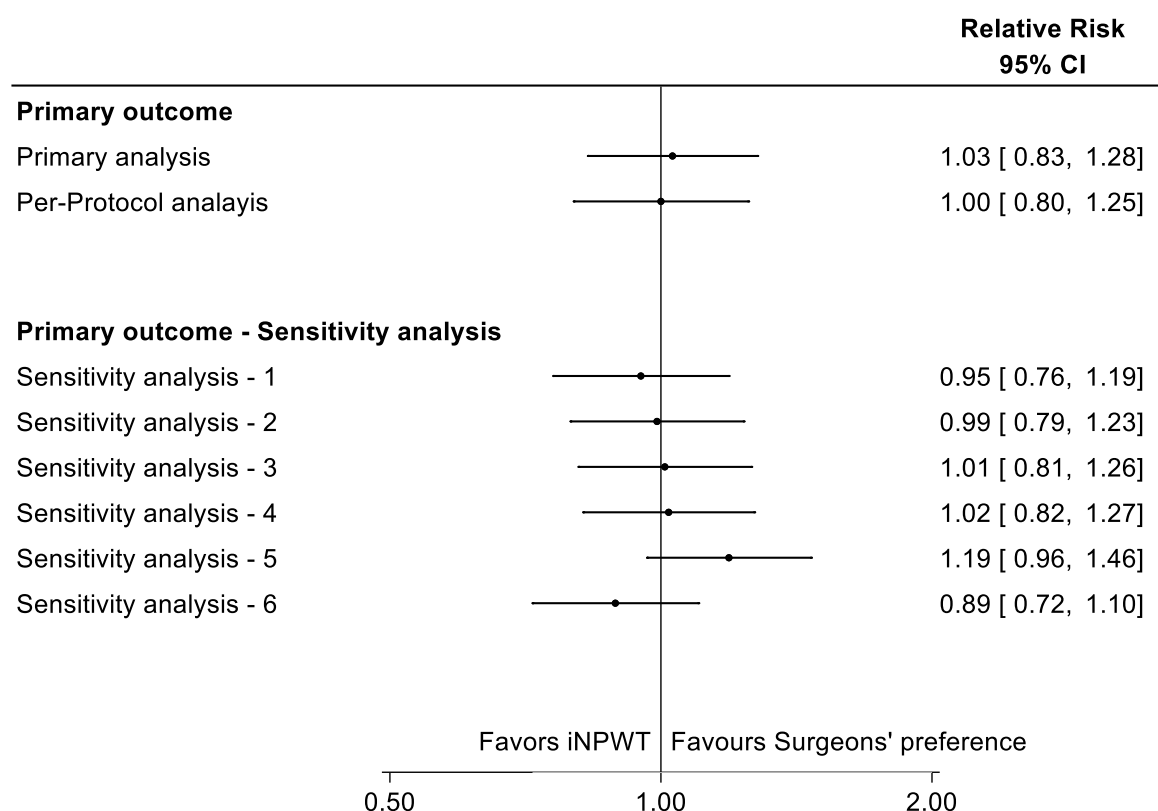
Supplementary eFigure 2. Definition of CDC criteria

SSI within 30 days of emergency laparotomy, as defined by the internationally accredited Centers for Disease Control (CDC) criteria

The infection must occur within 30-days of the index operation

AND at least one of:

- Purulent drainage from the wound
 - Organisms are detected from a wound swab
 - Wound opened spontaneously or by a clinician
- AND** At the surgical wound, the patient has at least one of:
- pain or tenderness
 - redness
 - heat
 - localised swelling
 - systemic fever (>38°C)
- Diagnosis of SSI by a clinician or on imaging

Supplementary eFigure 3. Sensitivity analysis forest plot**Sensitivity analysis 1:**

Analysis of the ITT population and excluding any 30-day forms completed before day 30 post-surgery but includes any 30-day forms completed after day 30

Sensitivity analysis 2:

Analysis of the ITT population and only including 30-day forms completed within 30-44 days (i.e. +14 day time-window) post-surgery

Sensitivity analysis 3:

Analysis of the ITT population excluding any SSI's identified from the patient diary only

Sensitivity analysis 4:

Analysis of the ITT population excluding any participants that return to theatre where the participant has not already had an SSI diagnosed

Sensitivity analysis 5:

Analysis of the ITT analysis population with missing data imputed as

- SSI (worst case) for the iNPWT group and No SSI (best case) for control group

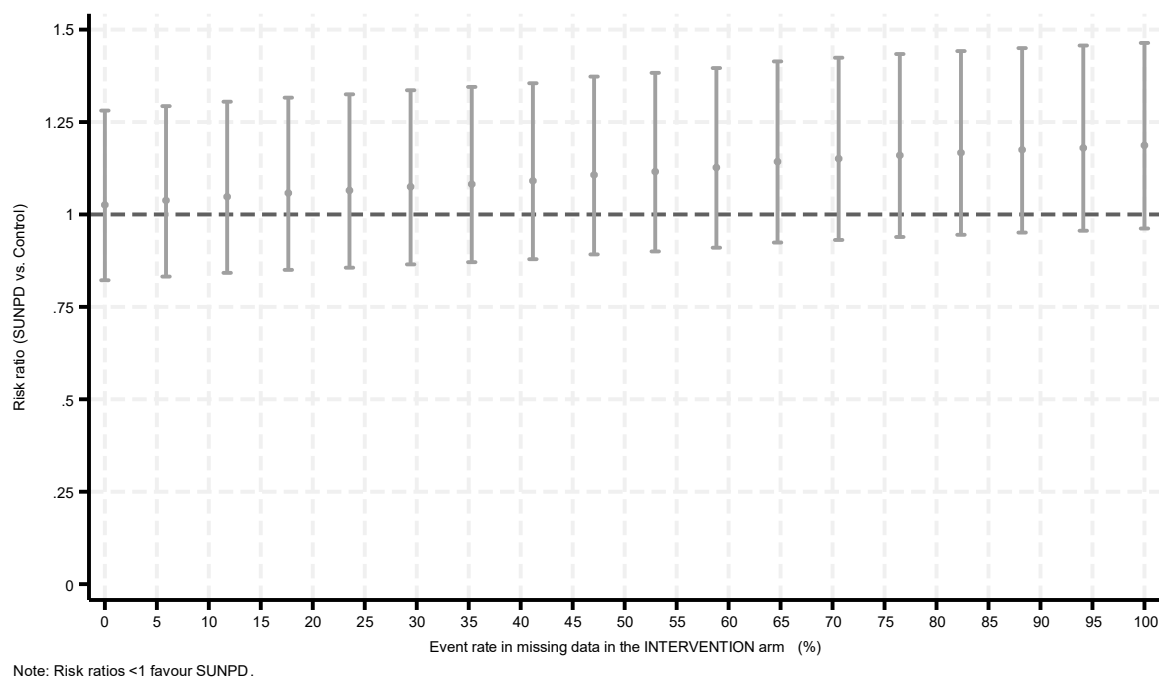
Sensitivity analysis 6:

Analysis of the ITT analysis population with missing data imputed as

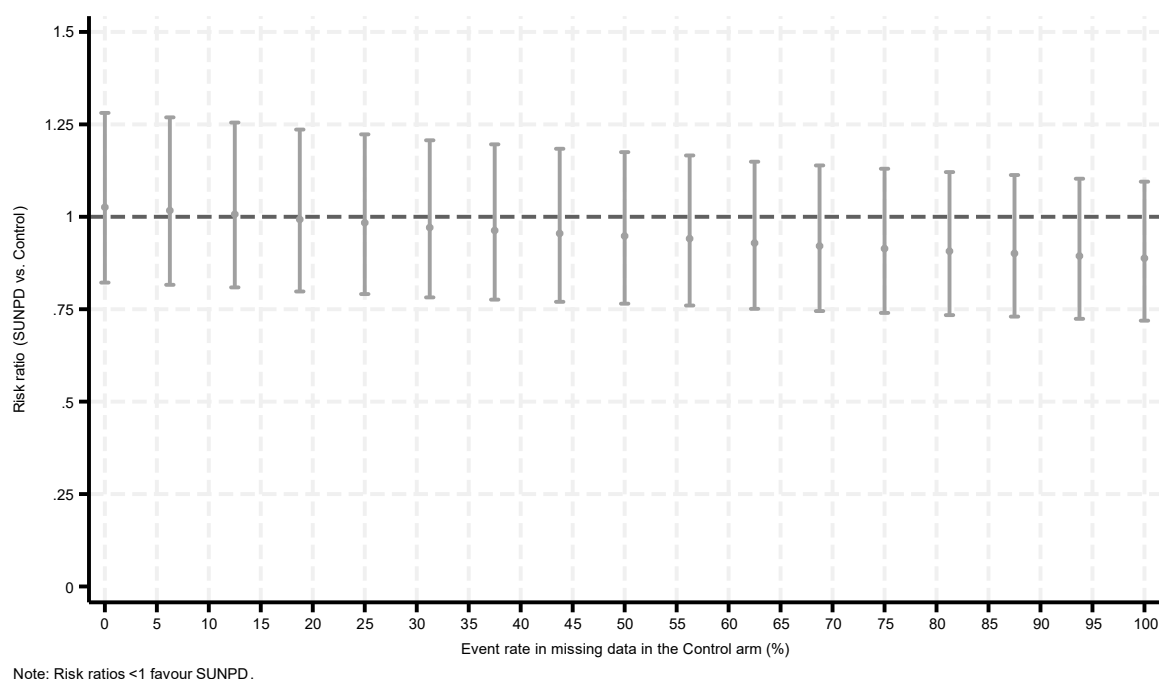
No SSI (best case) for the iNPWT group and SSI (worst case) for control group

Supplementary eFigure 4. Tipping point analysis

Graph A



Graph B



Graph A demonstrates that even if all missing data are imputed as SSI in the SUNPD group and no SSI in the control group, this will still not provide a result that statistically favours the control group.

Similarly graph B demonstrates that even if all missing data are imputed as SSI in the control group and no SSI in the SUNPD group, this will still not provide a result that statistically favours the SUNPD group.